Equations and Laws

Ohms Law:

$V=IR$

Power: $P=VI$ or $P=I^2R$

Series and Parallel circuits:

Series: $R_T = R_1 + R_2 + \dots$

Parallel: $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

Series:
- Same current flows through every component
- Voltage is divided across components

Parallel:
- each branch has the same voltage
- Current divides between branches

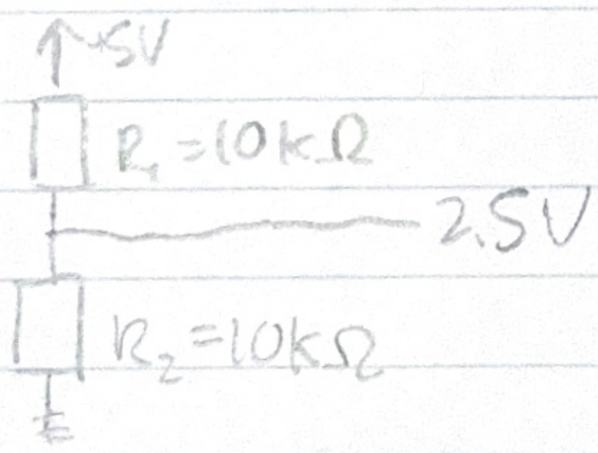
Kirchhoff's laws:

$\sum I_{in} = \sum I_{out}$

$\sum V = 0$ The sum of voltage rises and drops around any closed loop $= 0$

Voltage dividers:

$$V_{out} = V_{in} \frac{R_2}{R_1 + R_2}$$



$V_{out} = 2.5V$

- Equal resistors divide the input in half
- Inc the lower resistor inc the output voltage
- Inc the upper resistor dec the output voltage

Comparator behaviors:
High output when $V_+ > V_-$
Low output when $V_+ < V_-$
When at switching threshold, the output may be unstable

Pull-up and pull-down resistors:
PU: Connected between a signal and the positive supply
- Establishes a High voltage when nothing is actively pulling the signal low
- Prevents signal from floating
PD: Connected between signal and ground
- Establishes a low voltage
- Prevents a disconnected high-impedance digital input from floating
Floating inputs: No defined voltage

Logic Gates:
1 = High
0 = Low

AND: High when both inputs are High
OR: High when 1+ inputs are High
NOT: Inverts a signal
NAND: Opposite of ~~AND~~
NOR: Opposite of OR
* Inputs would be hard breaking detected and high current detected. Output would be both conditions detected.